

Comprehensive Evaluation of CNN Backbones and Self-Supervised Learning for PRBD Rice Classification

Abstract—Microscopic rice variety classification is important for agriculture, food safety and quality control, as correct identification helps ensure proper distribution, quality management and better crop planning. However this task is challenging because different rice varieties often look very similar and microscopic images contain very fine texture patterns that are difficult to separate. Manual labeling of these images is also time consuming, expensive and requires expert knowledge which makes fully supervised methods less practical. To address these challenges this study explores self-supervised learning (SSL) approaches to learn useful feature representations from unlabeled rice images. We investigate three SSL methods: contrastive learning (SimCLR, BYOL) and non-contrastive learning (MAE) using two different backbones including DenseNet121 and ViT. The pretrained features are evaluated using linear probing, neural network classifiers, traditional machine learning models and full fine-tuning. Our results show that SSL can learn strong representations even with limited labeled data. SimCLR and BYOL effectively capture various features between varieties while MAE excels at learning fine-grained textures. Overall the study demonstrates that SSL methods combined with appropriate backbones and fine-tuning strategies provide a practical and effective solution for microscopic rice variety classification reducing the need for large labeled datasets and supporting real-world agricultural applications.

Index Terms—microscopic rice classification, self-supervised learning, feature representation, agricultural image analysis

I. INTRODUCTION

Microscopic rice variety classification is significant for agriculture, food safety and quality control because correct identification helps proper distribution, unique identification, and better raising decisions. However this task is difficult because of the high visual similarity between different rice varieties and the presence of very fine texture patterns in microscopic images. Also manual labeling of microscopic rice images is time consuming, costly and requires expert knowledge which makes fully supervised learning Methods less practical. Supervised convolutional neural networks (CNNs) have been used in earlier research to classify rice images. These techniques can perform well but they generally fail to differentiate visually similar rice varieties with small texture changes and require a lot of labelled data [4]. Self supervised learning (SSL) which may extract valuable feature representations from unlabelled data, is motivated by these constraints. In recent years, strong performance of various self-supervised techniques such as SimCLR [1], BYOL [2] and Masked Autoencoders (MAE) [3] on learning image representations from natural images has been demonstrated. However, their

performance on detailed/textured rice microscopic image data has been less studied. Additionally, the impact of various CNN backbones on the feature learning performed by self-supervised techniques on Rice Microscopic images has yet to be clearly determined. The goal of this work is to identify the limitations mentioned above with a thorough and connected approach. Our main goals are to:

- Present the first self supervised learning study on the PRBD microscopic rice dataset for microscopic rice variety classification.
- SimCLR and BYOL learn discriminative features while MAE captures fine grained textures showing strengths in self-supervised texture representation.
- Evaluate SSL models using multiple CNN backbones including NASNetMobile, ConvNeXt-Tiny, InceptionV3, ResNet50 and DenseNet121 for study validity across different model architectures.
- Evaluate the feature transferability of linear probing, neural classifiers, and traditional machine learning classifiers and provide an analysis of the performance of SSL in limited labeled data and fine-tuning solutions that can be used in real world agricultural and food quality applications.

II. RELATED WORK

A. Conventional Machine Learning and Traditional Classification Methods

Although no models or accuracies were reported, Bhat-tacharjee et al. [5] provided a high resolution microscopic data set of 10 varieties of Bangladeshi rice (2,010 images, increased to 8,010) to support research on rice classification. Saxena et al. [6] applied traditional ML classifiers to handcrafted morphological and color features where Random Forest achieved 99.85% precision and Decision Tree 99.68% for five rice varieties. Using SVM on geometric and morphological features, Gopalakrishnan and Reagan [7] achieved 97.8% accuracy across various rice varieties. The success of feature based classification was confirmed by other studies using multi class SVM on shape and colour descriptors which reported 92.22% accuracy [8]. Li et al. [9] achieved 99.89% accuracy when combining NIR hyperspectral imaging with SVM, and traditional texture-based classifiers also performed well when grading rice quality

B. CNN-Based Deep Learning in Rice and Crop Classification

Various studies have explored CNN-based deep learning approaches for rice classification and agricultural image analysis tasks. Lin et al. introduced RiceNet, a deep convolutional neural network for rice variety classification using a large dataset of 75,000 grain images, achieving up to 94–100% accuracy and outperforming traditional architectures such as InceptionV3 and ResNetInceptionV2 in this task [10]. Another study combined hyperspectral imaging with a CNN-Transformer hybrid, achieving 95.33% accuracy in classifying hybrid Indica paddy rice varieties. Additionally, multi-dimensional fusion CNNs, such as MDDF-CNN and MDFF-CNN, improved classification accuracy to over 90% using near-infrared hyperspectral data.

C. Self-Supervised Learning (SSL) for Agricultural Vision

Self-supervised learning (SSL) is very useful for rice classification because it can learn from data without labels. Petchhnan and Su used SSL methods like SimCLR and MoCo to classify moving rice grains. Their method improved accuracy by 7.5% in tests and 2.4% in real situations. Gao et al. [14] used a masked autoencoder with SegFormer to analyze rice field images and got better results than normal supervised models. These studies show that SSL can learn important features like shape and texture, making it helpful for rice analysis when labeled data is limited.

D. Limitations and Research Gap

Previous studies on rice classification show good results but have some problems. Traditional machine learning methods [5]–[9] use handmade features like texture, shape and color which makes it difficult to use them for new rice varieties. CNN-based models [10]–[12] like VGG, ResNet or Inception give high accuracy (90–100%) but need large labeled datasets and are difficult to understand. Self-supervised learning (SSL) methods [13]–[14] such as contrastive learning (SimCLR, BYOL) and non-contrastive learning (MAE) reduce the need for labels but very few studies have used them for rice data. Also modern CNN models like ConvNeXt have not been combined with SSL and explainable AI (XAI) in previous works. There is still a need for a strong and easy-to-understand system that uses high-resolution rice data, modern CNNs, SSL methods and XAI to classify rice types accurately and in a way that people can understand. Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads— $\text{\LaTeX}$ will do that for you.

III. METHODOLOGY

To classify different rice types in the PRBD dataset, we evaluate both supervised and self-supervised learning (SSL) methods. Our approach assesses the effectiveness of various CNN models and SSL techniques, even with limited labeled data. Initially, we clean and augment the dataset. In supervised learning, models are trained with labeled images, while in SSL, we first train the model without labels to learn features before testing their classification performance.

A. Dataset description:

This study uses the Processed Rice Varieties of Bangladesh (PRBD) dataset, which contains microscopic images of 10 processed rice varieties collected from local markets in Bangladesh. The dataset includes a total of 2,000 original microscopic images, with 200 images per class. To address data scarcity and class imbalance, an additional 8,000 augmented images are provided, resulting in 800 augmented images per class.



Fig-2: Sample images of PRBD Rice Varieties in BD

- Aush: 200 original
- Beroi: 200 original
- BR-28: 200 original
- BR-29: 200 original
- Ghee Bhog: 200 original
- Katari Nazir: 200 original
- Katari Siddho: 200 original
- Swarna: 200 original
- Miniket: 200 original
- Chinigura: 200 original

B. Equations and Loss Functions

Self-Supervised Learning (SSL) Losses

1) SIMCLR (Contrastive Learning): NT-Xent Loss:

$$\mathcal{L}_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

Where:

- z_i, z_j are the normalized feature vectors of two augmented views of the same image.

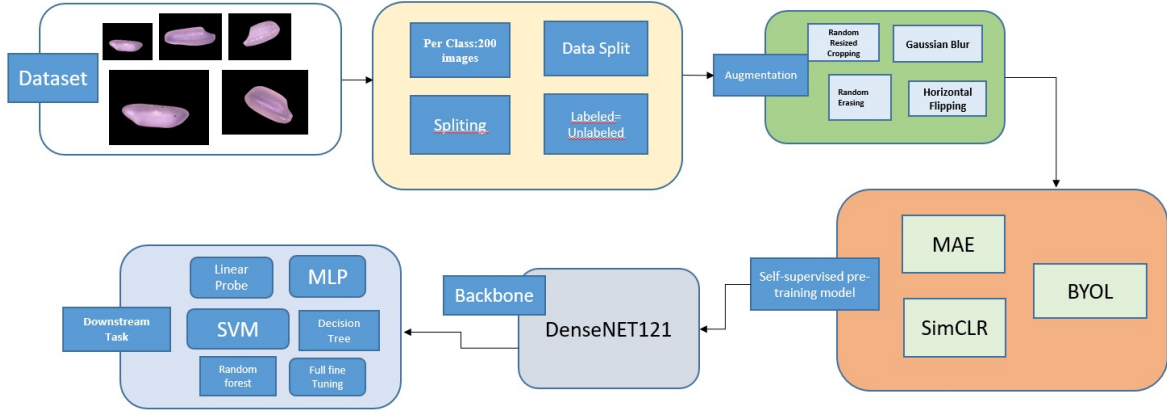


Fig. 1. Overview of the Research Methodology

- τ is the temperature parameter, controlling the softness of probabilities.
- N is the batch size.

2) *BYOL (Bootstrap Your Own Latent): Negative Cosine Similarity Loss:*

$$\mathcal{L}_{\text{BYOL}} = 2 - 2 \cdot \frac{p_{\theta}(z_i) \cdot z'_t}{\|p_{\theta}(z_i)\| \|z'_t\|}$$

Where:

- $p_{\theta}(z_i)$ is the predictor output from the online network.
- z'_t is the target network embedding (EMA updated).

3) *MAE (Masked Autoencoder): Reconstruction (MSE) Loss:*

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|^2$$

Where:

- $\mathcal{M}$ is the set of masked patches.
- x_i is the original patch.
- $\hat{x}_i$ is the reconstructed patch from the encoder-decoder.

4) *Supervised Learning Losses: Cross-Entropy Loss (Classification)*

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

Where:

- C is the number of rice classes.
- y_c is the ground truth (one-hot encoded).
- $\hat{y}_c$ is the predicted probability for class c .

5) *Evaluation Metrics: Accuracy*

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total predictions}}$$

6) *Precision, Recall:*

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

Where:

- TP : True Positives
- FP : False Positives
- FN : False Negatives

C. Model Description

1) *SIMCLR(Simple Framework for Contrastive Learning of Visual Representations):* SIMCLR is a self-supervised learning model that uses DenseNet121 as its backbone. The original classifier layer of DenseNet121 is removed and replaced with a projection head that transforms image features into a compact latent space. For each rice image in the PRBD microscopic dataset, two augmented versions are generated using standard transformations such as random cropping, flipping, color changes, and blurring. The model is trained with contrastive learning, which encourages the features of the two augmented views of the same image to be close together in feature space, while pushing the features of different images apart. On the rice dataset SIMCLR focuses on learning meaningful feature representations without using any labels. After pretraining, the learned features can be used for downstream tasks like classifying rice varieties. The pretrained encoder can either be frozen and used with simple classifiers such as linear probe, MLP, SVM, or Random Forest, or it can be fully fine-tuned together with a classifier for improved performance. This approach allows the model to capture subtle differences in rice texture, shape, and appearance, which are important for fine-grained classification.

2) *BYOL(Bootstrap Your Own Latent):* BYOL (Bootstrap Your Own Latent) is another self-supervised learning method that also uses DenseNet121 as its backbone. Similar to SIMCLR, the classifier layer is removed, and the model uses a combination of a projection head and a prediction head. Each

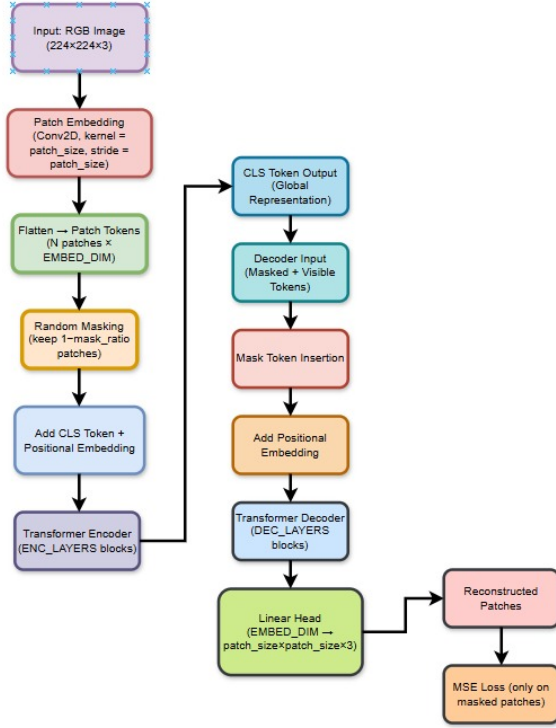


Fig. 2. MAE Model Architecture

image is augmented twice, and the model is trained using negative cosine similarity. The key difference from SIMCLR is that BYOL does not require negative pairs. Instead, it uses an exponentially moving average (EMA) target network to guide the learning of the online network. In the PRBD rice dataset, BYOL focuses on learning high-quality feature representations of microscopic rice images. These features capture intricate patterns, textures, and structural information of the rice grains. After pretraining, the learned encoder can be applied to downstream classification tasks. Similar to SIMCLR, the encoder can be frozen and combined with various classifiers or fully fine-tuned together with a classifier for optimal performance. BYOL's design allows it to extract rich and robust representations even from unlabeled data, making it highly effective for fine-grained classification tasks.

3) *MAE (Masked Autoencoder)*: MAE uses a Vision Transformer (ViT) as its backbone for self-supervised learning. Unlike contrastive methods, MAE works by masking a large portion of image patches 75/ and training the model to reconstruct the missing patches using the visible ones. This encourages the model to learn global and local structural information from each image. The pretraining is performed only on the encoder-decoder structure, and after pretraining the decoder is removed, leaving a pretrained encoder that contains rich image representations. On the PRBD microscopic rice dataset MAE learns detailed contextual features that capture the internal structure and texture of rice grains. These features can then be used for downstream rice variety classification.

The pretrained encoder can either be frozen and used with simple classifiers or it can be fully fine-tuned together with a classifier for improved accuracy. MAE's approach is especially useful for learning features from high-resolution images where the structure of small image patches carries significant information.

4) *Pretrained Supervised Deep Learning Models*: Several pretrained supervised deep learning models are used for rice variety classification. These models are initially trained on large datasets like ImageNet, which allows them to learn general visual features. They are then fine-tuned on the PRBD microscopic rice images so that these features can adapt to the unique textures, shapes and structures of rice grains. NASNetMobile is a lightweight CNN designed with Neural Architecture Search (NAS) to balance accuracy and efficiency. It acts as both a feature extractor and classifier, using pre-trained weights to identify patterns in rice grain texture and structure. ConvNeXt-Tiny is a modern convolutional network inspired by Vision Transformers, incorporating large kernels and efficient normalization to capture subtle variations in grain appearance. InceptionV3 uses inception modules to process image features at multiple scales, making it suitable for rice grains of different sizes and shapes. ResNet50 uses residual connections to learn robust deep representations of textures and edges, while DenseNet121 connects each layer to all previous layers, promoting feature reuse and capturing fine-grained details in rice grains. Together these pretrained models leverage knowledge from large-scale datasets while adapting to the PRBD rice dataset through fine-tuning. This combination allows them to extract meaningful features from rice images and perform accurate fine-grained classification of different rice varieties.

IV. RESULT AND DISCUSSION

A. Performance of Pretrained Supervised Models

This section presents a detailed comparative analysis of five supervised CNN models NASNetMobile, ConvNeXt-Tiny, InceptionV3, ResNet50, and DenseNet121 evaluated on the PRBD microscopic rice dataset using different train-test split ratios.

NASNetMobile demonstrates strong and stable performance across most splits. The model achieves an accuracy of 0.9033 at the 70:30 split with precision 0.9049, recall 0.9033, and F1-score 0.9029. Its best performance is obtained at the 30:70 split, where accuracy reaches 0.9321, precision 0.9327, recall 0.9322, and F1-score 0.9322. However, performance degrades at extreme splits, dropping to 0.7839 accuracy at 10:90, indicating sensitivity to very limited training data. ConvNeXt-Tiny shows moderate but inconsistent performance. At the 60:40 split, it achieves an accuracy of 0.8812 with precision 0.8981 and F1-score 0.8794, which is its best-performing configuration. Balanced splits such as 50:50 and 40:60 also yield reasonable accuracies of 0.8700 and 0.8600, respectively. However, a significant performance drop is observed at the 30:70 split, where accuracy falls to 0.5914, highlighting instability under data imbalance.

TABLE I
PERFORMANCE METRICS OF DIFFERENT MODELS WITH VARIOUS
TRAIN:TEST SPLITS

Model	Train:Test	Accuracy	Precision	Recall	F1-Score
NASNetMobile	90:10	0.8650	0.8687	0.8650	0.8608
	80:20	0.8875	0.8896	0.8875	0.8876
	70:30	0.9033	0.9049	0.9033	0.9029
	60:40	0.8950	0.8982	0.8950	0.8948
	50:50	0.8820	0.8828	0.8820	0.8817
	40:60	0.8860	0.8876	0.8867	0.8868
	30:70	0.9321	0.9327	0.9322	0.9322
	20:80	0.8313	0.8354	0.8313	0.8327
	10:90	0.7839	0.7839	0.7839	0.7830
ConvNeXtTiny	90:10	0.8200	0.8349	0.8200	0.8149
	80:20	0.8175	0.8203	0.8175	0.8154
	70:30	0.7450	0.7533	0.7450	0.7393
	60:40	0.8812	0.8981	0.8812	0.8794
	50:50	0.8700	0.8907	0.8700	0.8690
	40:60	0.8600	0.8822	0.8600	0.8595
	30:70	0.5914	0.6124	0.5914	0.5625
	20:80	0.8062	0.8486	0.8063	0.8067
	10:90	0.7911	0.8163	0.7911	0.7836
InceptionV3	90:10	0.9150	0.9209	0.9150	0.9142
	80:20	0.9200	0.9213	0.9200	0.9203
	70:30	0.9067	0.9111	0.9417	0.7396
	60:40	0.8988	0.9073	0.8988	0.8983
	50:50	0.8650	0.8753	0.8650	0.8666
	40:60	0.8775	0.8861	0.8775	0.8774
	30:70	0.8648	0.8778	0.8651	0.8635
	20:80	0.8025	0.8372	0.8025	0.8044
	10:90	0.7956	0.8131	0.7956	0.7950
ResNet50	90:10	0.7300	0.7341	0.7300	0.7259
	80:20	0.7475	0.7566	0.7475	0.7421
	70:30	0.7417	0.7511	0.7417	0.7396
	60:40	0.6875	0.7487	0.6875	0.6804
	50:50	0.6960	0.7557	0.6960	0.6841
	40:60	0.6758	0.7370	0.6758	0.6679
	30:70	0.4553	0.4686	0.4562	0.4193
	20:80	0.6400	0.6931	0.6400	0.6243
	10:90	0.5778	0.6060	0.5778	0.5617
DenseNet121	90:10	0.8900	0.9395	0.9350	0.9345
	80:20	0.9450	0.9463	0.9450	0.9449
	70:30	0.9283	0.9290	0.9283	0.9282
	60:40	0.9163	0.9163	0.9163	0.9159
	50:50	0.9150	0.9183	0.9150	0.9155
	40:60	0.9033	0.9059	0.9033	0.9039
	30:70	0.9470	0.9478	0.9470	0.9471
	20:80	0.8788	0.8798	0.8787	0.8784
	10:90	0.8483	0.8494	0.8483	0.8455

InceptionV3 performs well when sufficient training data is available. At the 80:20 split, it achieves an accuracy of 0.9200, precision 0.9213, recall 0.9200, and F1-score 0.9203, while the 90:10 split also produces high accuracy of 0.9150. Performance gradually decreases as the training data is reduced, with accuracy dropping to 0.7956 at the 10:90 split. This trend indicates that InceptionV3 benefits significantly from larger labeled datasets.

ResNet50 shows comparatively weaker performance across all splits. The highest accuracy of 0.7475 is achieved at the 80:20 split, with precision 0.7566 and F1-score 0.7421. Performance deteriorates substantially under imbalanced splits, reaching only 0.4553 accuracy at the 30:70 split and 0.5778 at 10:90, suggesting limited effectiveness in capturing fine-grained microscopic texture features for this task. DenseNet121 consistently outperforms all other models across nearly all split ratios. At the 80:20 split, it achieves an accuracy of 0.9450, precision 0.9463, recall 0.9450, and F1-score 0.9449. The highest performance is observed at the 30:70 split, where accuracy reaches 0.9470 with precision 0.9478,

recall 0.9470, and F1-score 0.9471. Even under extreme splits such as 10:90, DenseNet121 maintains strong performance with an accuracy of 0.8483, demonstrating superior robustness and generalization capability.

Overall the results clearly indicate that DenseNet121 is the most effective architecture for microscopic rice variety classification, followed by NASNetMobile and InceptionV3. ConvNeXt-Tiny shows moderate but unstable behavior, while ResNet50 performs the weakest. These findings highlight the importance of dense feature reuse and strong representation learning for fine-grained, texture-rich microscopic image classification tasks.

TABLE II
PERFORMANCE COMPARISON OF SELF-SUPERVISED LEARNING MODELS

Method	SimCLR	BYOL	MAE
Backbone	DenseNet121	DenseNet121	ViT
SVM	0.77	0.91	0.87
Linear Probe	0.82	0.92	0.90
MLP	0.81	0.92	0.89
Decision Tree	0.67	0.82	0.65
Random Forest	0.81	0.90	0.82
Fine Tuning	0.92	0.96	0.98

B. Performance Analysis of Self-Supervised Learning Models

The performance analysis shows that BYOL is the most consistent model, achieving high accuracy across almost all classifiers. While SimCLR performs well, BYOL outperforms it in every test, particularly with classical tools like SVM and Random Forest. This suggests that BYOL's features are more robust and easier for different classifiers to understand.

MAE stands out when using Fine Tuning, reaching the highest overall accuracy of 0.98. This shows that the Vision Transformer (ViT) backbone is extremely powerful for rice image recognition when fully adjusted. In contrast, simpler methods like Linear Probe and MLP also provide strong results, proving that all three models successfully learn meaningful patterns from the dataset. However traditional models like Decision Trees performed poorly compared to others. This indicates that self-supervised features are complex and work best with more advanced classifiers or direct fine-tuning. Overall, BYOL is the most reliable choice for general use, while MAE is the best for maximum accuracy through fine-tuning.

CONCLUSION

This study presents a comprehensive self-supervised learning (SSL) framework for microscopic rice variety classification using the PRBD dataset. By leveraging unlabeled data, the proposed approach addresses the limitations of traditional machine learning and fully supervised deep learning methods, which rely heavily on handcrafted features or large annotated datasets. Three SSL paradigms SimCLR, BYOL, and MAE were systematically evaluated using modern backbones and multiple downstream classifiers to assess their representation learning capability. Experimental results demonstrate that SSL

significantly enhances classification performance when combined with appropriate downstream strategies. Among the evaluated methods, BYOL with a DenseNet121 backbone provides the most consistent performance across different classifiers, achieving up to 0.96 accuracy through fine-tuning. MAE, based on a Vision Transformer backbone, achieves the highest overall accuracy of 0.98 after fine-tuning, highlighting the strong potential of reconstruction-based SSL models for fine-grained microscopic image analysis. SimCLR also shows notable improvement when fine-tuned, reaching an accuracy of 0.92, despite relatively weaker performance with traditional classifiers. A comparative analysis with existing literature confirms that while traditional machine learning and supervised CNN-based models can achieve high accuracy, they lack scalability and depend on extensive labeled data or handcrafted features. In contrast, the proposed SSL-based framework learns robust and transferable representations directly from unlabeled images, making it more suitable for real-world agricultural applications where labeled data is scarce. Overall, this work demonstrates that self-supervised learning, particularly BYOL and MAE, offers an effective and label-efficient solution for microscopic rice variety classification. The proposed framework can be extended to other crop types and agricultural imaging tasks, and future work may explore larger-scale datasets, hybrid SSL strategies and explainable AI techniques to further enhance model transparency and practical deployment.

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